

ECO 362 Financial Investments

Precept 6: Stocks

October 27 – October 31, 2025

1 Gordon Growth Model

2 Time-Series Tests of Return Predictability

3 Cross-Sectional Tests of Return Predictability

- An asset is paying dividends
- Let D_t denote dividend and P_t denote price (after dividend payment),
- Return from t to $t + 1$ is a random variable R_{t+1}

$$1 + R_{t+1} \equiv \frac{D_{t+1} + P_{t+1}}{P_t} \quad (1)$$

- **Assumption 1:** Expected returns are constant

$$\mathbb{E}_t[R_{t+1}] = \bar{R}, \quad \forall t$$

- Take expectations $\mathbb{E}_t$ of (1) and rearrange:

$$P_t = \frac{\mathbb{E}_t[D_{t+1}]}{1 + \bar{R}} + \frac{\mathbb{E}_t[P_{t+1}]}{1 + \bar{R}} \quad (2)$$

- Equation (2) applies for any time period:

$$P_{t+1} = \frac{\mathbb{E}_{t+1}[D_{t+2}]}{1 + \bar{R}} + \frac{\mathbb{E}_{t+1}[P_{t+2}]}{1 + \bar{R}}$$

- Iterate backward to period 0 (now) and apply law of iterated expectations:

$$\begin{aligned} P_0 &= \frac{\mathbb{E}_0[D_1]}{1 + \bar{R}} + \frac{\mathbb{E}_0[P_1]}{1 + \bar{R}} \\ &= \frac{\mathbb{E}_0[D_1]}{1 + \bar{R}} + \frac{\mathbb{E}_0\left[\frac{\mathbb{E}_1[D_2]}{1 + \bar{R}} + \frac{\mathbb{E}_1[P_2]}{1 + \bar{R}}\right]}{1 + \bar{R}} \\ &= \frac{\mathbb{E}_0[D_1]}{1 + \bar{R}} + \frac{\mathbb{E}_0[D_2]}{(1 + \bar{R})^2} + \frac{\mathbb{E}_0[P_2]}{(1 + \bar{R})^2} \\ &\dots \\ &= \sum_{t=1}^{\infty} \frac{\mathbb{E}_0[D_t]}{(1 + \bar{R})^t} \end{aligned}$$

- Interpretation: Price is the discounted value of expected dividends

- **Assumption 2:** expected dividends grow at constant rate $G < \bar{R}$ per period:

$$\mathbb{E}_0[D_t] = D_0(1 + G)^t$$

- Then the dividend discount model tells us

$$\begin{aligned} P_0 &= \sum_{t=1}^{\infty} \frac{D_0(1 + G)^t}{(1 + \bar{R})^t} \\ &= D_0 \sum_{t=1}^{\infty} \left(\frac{1 + G}{1 + \bar{R}} \right)^t \\ &= D_0 \frac{\frac{1+G}{1+\bar{R}}}{1 - \frac{1+G}{1+\bar{R}}} \\ \Rightarrow \frac{P_0}{D_0} &= \frac{1 + G}{\bar{R} - G} \end{aligned}$$

$$\frac{P_0}{D_0} = \frac{1 + G}{\bar{R} - G}$$

- How does price-dividend ratio relate to G ? to $\bar{R}$?
- Suppose that G is constant.
 - If price-dividend ratio is high, what does that say about $\bar{R}$?
 - If price-dividend ratio is low, what does that say about $\bar{R}$?

$$\frac{P_0}{D_0} = \frac{1 + G}{\bar{R} - G}$$

- Historical examples of when ratio was high? was low?
 - High: the dot-com bubble
 - Low: the 2008 financial crisis
- Why is the ratio higher since 1980s?
 - Firms paid fewer cash dividends;
 - Investors are less risk averse.
- Dividends may no longer be a good signal of fundamental value.
 - Need to look at earnings or adjusted dividends that incorporate share repurchases.

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Return Predictability: Time-Series Regression

Question: Can we find some time series variable x_t that predicts future excess returns?

Method: Run the regression $r_{s,t+1} - r_{b,t+1} = \delta_0 + \delta_1 x_t + \epsilon_{t+1}$.

- $r_{s,t+1} - r_{b,t+1}$: realized excess return in year $t + 1$
- x_t : predictor variable at the end of year t , e.g., *price-earning ratio*.

Example: Example 1 in “Stock Examples”, table titled “P/E10”.

Interpreting Results:

- Hypothesis: $H_0 : \delta_1 = 0$ vs $H_1 : \delta_1 \neq 0$.
- Regression result from 1920–2020 data: $\hat{\delta}_1 = -0.0053$, $[t = -2.24]$.
 - Reject the null hypothesis at 5% significance level
 - High price-to-earning ratios predict low excess return in the next year.

- Expected excess returns vary with price-to-earning ratio x_t :

$$\mathbb{E}_t[r_{s,t+1} - r_{b,t+1}] = \hat{\delta}_0 + \hat{\delta}_1 x_t$$

- Therefore optimal portfolio varies with x_t :

$$w_t = \frac{\mathbb{E}_t[r_{s,t+1} - r_{b,t+1}]}{\gamma \sigma^2} = \frac{\hat{\delta}_0 + \hat{\delta}_1 x_t}{\gamma \sigma^2}$$

- E.g., if $\hat{\delta}_1 < 0$, then when price-earning is high, hold less stocks; when price-earning is low, buy more stocks
- See “Example 2” in Excel with $\gamma = 5$.

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- The dividend-price ratio for stock i according to the Gordon growth model.

$$\frac{D_i}{P_i} = \frac{\bar{R}_i - G_i}{1 + G_i} \quad (3)$$

- Empirically, stocks with high dividend-price ratios have high future returns.
- Value stocks: **high** dividend-price, earnings-price, or book-to-market equity;
- Growth stocks: **low** dividend-price, earnings-price, or book-to-market equity.

Return Predictability: Cross-Section Regression

- ❶ Find a stock-specific variable $x_{i,t}$, e.g., book-to-market ratio.
- ❷ Construct portfolios according to the following procedure:
 - At end of year t , sort stocks by $x_{i,t}$ into 10 decile portfolios.
 - Compute return of each portfolio in year $t + 1$: $r_{k,t+1}$ ($k = 1 \cdots 10$)
 - Re-sort by $x_{i,t+1}$ at the end of year $t + 1$, ...
- ❸ Use the 10 portfolios $r_{k,t}$ ($k = 1 \cdots 10$) to test CAPM.
 - For each $k = 1 \cdots 10$, run the CAPM regression
$$r_{k,t} - r_{b,t} = \alpha_k + \beta_k[r_{m,t} - r_{b,t}] + \varepsilon_{k,t}$$
 - Null hypothesis: $\alpha_k = 0$

Example: Example 3 in “Stock Examples”.